# AIML Assignment 10

**ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING**

**ASSIGNMENT – 10**

# Design a Model to Detect Fake News from Multimodal Data

**Dataset: Fakeddit Multimodal Dataset**

## 1. Aim

To design and implement a multimodal machine learning model for detecting fake news by combining textual and visual information from the Fakeddit dataset, and to evaluate the model using standard classification metrics.

## 2. Objectives

- To understand the concept of multimodal fake news detection.
- To preprocess textual and image-based information from a real-world dataset.
- To extract textual features using TF-IDF.
- To extract visual features using a pretrained ResNet50 model.
- To combine text and image features into a multimodal representation.
- To train a classification model for binary fake-news detection.
- To compare text-only, image-only and multimodal approaches using accuracy, precision, recall and F1-score.
- To analyze predictions using confusion matrices.

## 3. Dataset Description

The Fakeddit dataset is a multimodal fake-news dataset containing social-media posts with textual and visual information. For this experiment, the multimodal subset was used. The field **clean_title** was used as the text input, **image_url** was used to obtain the corresponding image, and **2_way_label** was used as the binary target.

| **Property** | **Value** |
|---|---|
| Dataset | Fakeddit Multimodal Dataset |
| Text feature | clean_title |
| Image feature | image_url |
| Target | 2_way_label |
| Classification | Binary |
| Text representation | TF-IDF |
| Image representation | Pretrained ResNet50 |
| Classifier | Logistic Regression |
| Training samples | 572 |
| Validation samples | 286 |
| Test samples | 286 |

## 4. Theory

**Multimodal learning** is a machine learning approach in which information from two or more modalities, such as text, images, audio or video, is combined to make a prediction. In fake news detection, relying only on text may

miss misleading or manipulated visual content, while an image-only model may ignore important claims expressed in the text.

**Text feature extraction using TF-IDF:** Term Frequency–Inverse Document Frequency converts text into numerical features. Term frequency measures how often a word occurs in a document, while inverse document frequency reduces the importance of words that occur in many documents. TF-IDF therefore gives higher importance to words that are relatively informative for a document.

**Visual feature extraction using ResNet50:** ResNet50 is a deep convolutional neural network with residual connections. A pretrained ResNet50 can be used as a feature extractor by removing its final classification layer and using the resulting feature vector as a representation of the image. This allows useful visual information to be extracted without training a complete CNN from scratch.

**Feature fusion:** After extracting text and image representations, the two feature vectors are concatenated to create a combined multimodal representation. The classifier then uses information from both modalities to predict the news class.

**Logistic Regression:** Logistic Regression is a supervised classification algorithm that estimates the probability of a class. It is suitable for this experiment because the fused representation is used for binary classification.

**Evaluation metrics:** Accuracy measures the proportion of correct predictions. Precision measures how many predicted positive cases are actually positive. Recall measures how many actual positive cases are correctly identified. F1-score is the harmonic mean of precision and recall and provides a balanced measure of classification performance.

**Confusion matrix:** A confusion matrix summarizes correct and incorrect predictions using true positives, true negatives, false positives and false negatives. It helps examine class-specific performance rather than relying only on overall accuracy.

## 5. Methodology

- Load the Fakeddit multimodal training, validation and test files.
- Select samples containing both text and image information and remove missing or invalid records.
- Create a balanced subset for practical execution in Google Colab.
- Use clean_title as the textual input and 2_way_label as the binary target.
- Transform the text into TF-IDF feature vectors.
- Download the corresponding images and preprocess them for ResNet50.
- Extract visual features using pretrained ResNet50.
- Scale the visual features and concatenate them with the TF-IDF features.
- Train Logistic Regression models using text-only, image-only and multimodal features.
- Evaluate the models using accuracy, precision, recall and F1-score.
- Generate class-distribution and confusion-matrix visualizations.

# 6. Outputs

## 6.1 Model Performance

| Model | Accuracy | Precision | Recall | F1-score |
|---|---|---|---|---|
| Text Only | 0.6713 | 0.7765 | 0.4681 | 0.5841 |
| Image Only | 0.7308 | 0.7319 | 0.7163 | 0.7240 |
| Multimodal | 0.7308 | 0.7319 | 0.7163 | 0.7240 |

## 6.2 Multimodal Classification Performance

The multimodal model achieved an accuracy of **73.08%**, precision of **0.7319**, recall of **0.7163**, and F1-score of **0.7240** on the test data. The Image-only model obtained the same scores in this run, while the Text-only model obtained 67.13% accuracy.

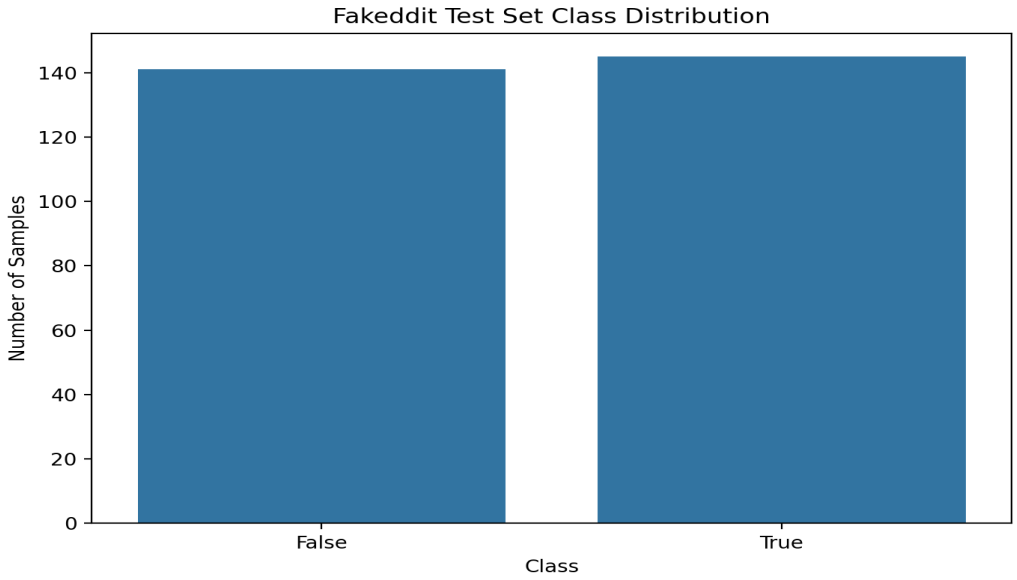



*Figure 1: Distribution of Fakeddit test classes*

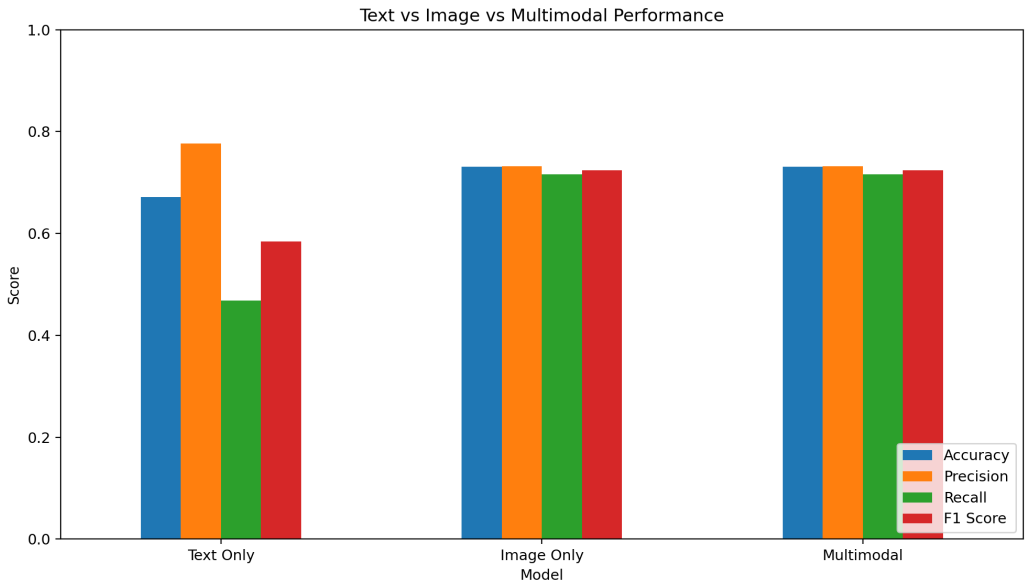



*Figure 2: Comparison of text-only, image-only and multimodal models*

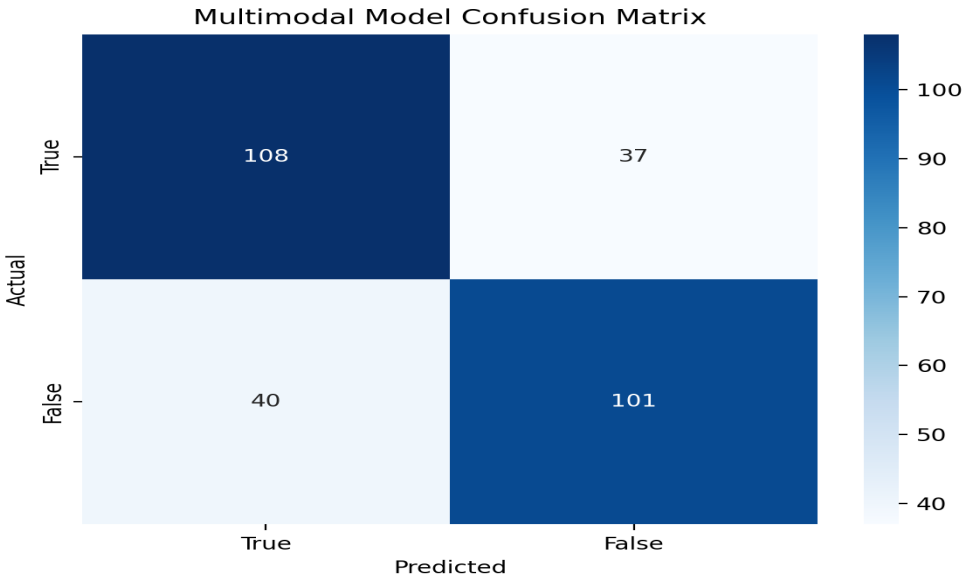



*Figure 3: Multimodal model confusion matrix*

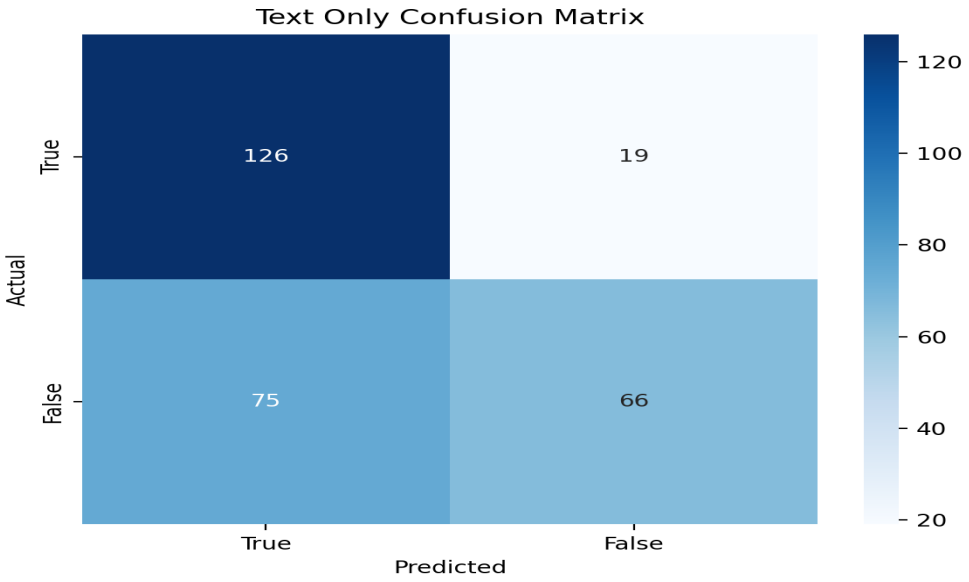



*Figure 4: Text-only model confusion matrix*

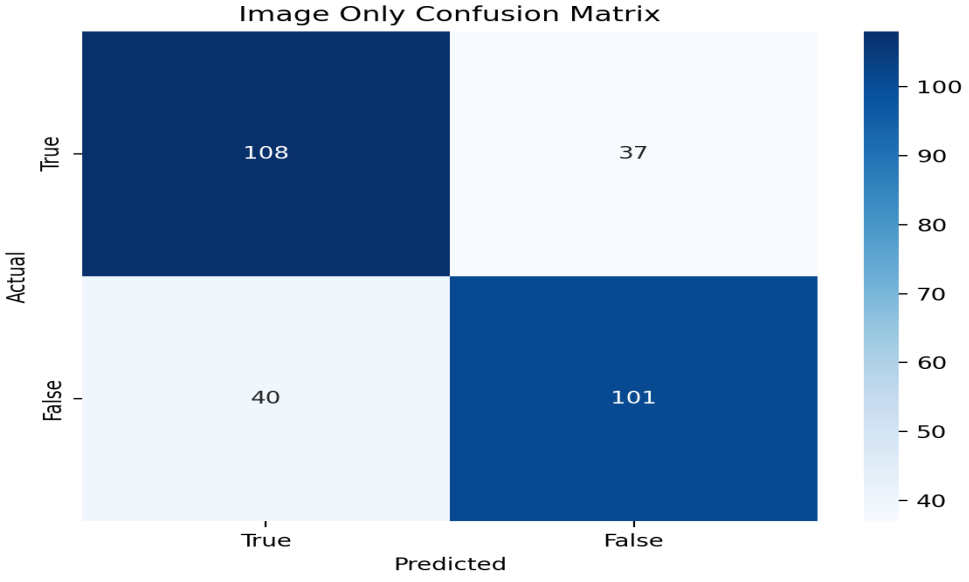



*Figure 5: Image-only model confusion matrix*

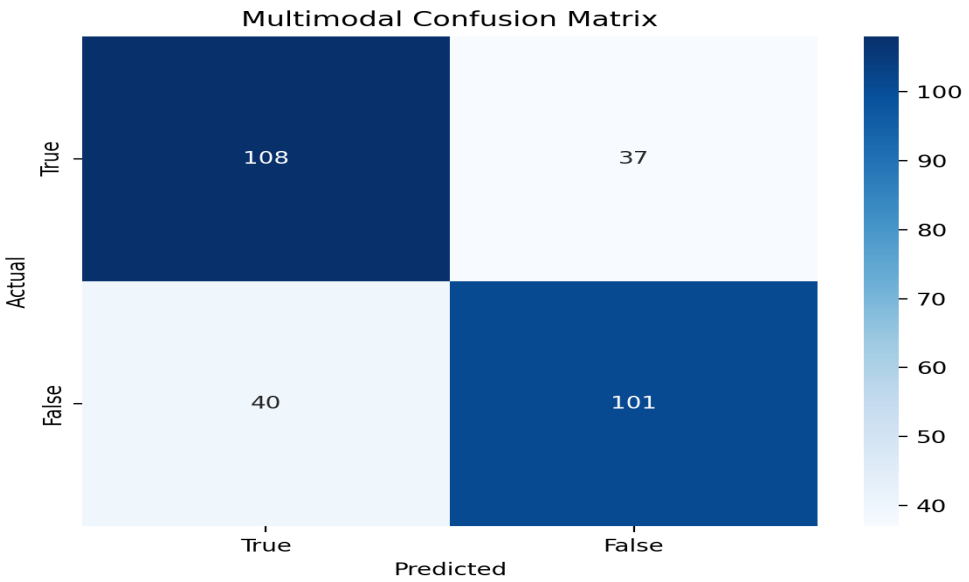



*Figure 6: Multimodal model confusion matrix*

## 7. Result

The multimodal fake news detection model was successfully implemented using the Fakeddit dataset. Textual information was converted into TF-IDF features and image information was represented using pretrained ResNet50 features. These representations were fused and classified using Logistic Regression. The multimodal model achieved **73.08% accuracy** and an **F1-score of 0.7240** on the test set.

## 8. Observations

- The experiment used both textual and visual information rather than depending on a single modality.
- The Text-only model achieved 67.13% accuracy, whereas the Image-only and Multimodal models achieved 73.08% accuracy in this run.
- The multimodal model obtained precision of 0.7319 and recall of 0.7163.
- The multimodal F1-score was 0.7240, indicating a balanced performance between precision and recall.
- The confusion matrix provides a clearer view of correctly and incorrectly classified samples for each class.
- The result shows that visual information can provide useful signals for fake-news classification.

## 9. Conclusion

The experiment demonstrates the application of multimodal machine learning to fake news detection. Textual features extracted using TF-IDF and visual features extracted using ResNet50 were combined to construct a multimodal representation. The resulting classifier successfully performed binary classification and was evaluated using multiple performance metrics and confusion matrices. The experiment shows that combining information from different modalities is a practical approach for analyzing online news content.